

Cross-species ncRNA annotation using synteny-constrained embedding similarity

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Abstract

Reference-based annotation transfer via orthology has been highly successful for protein-coding genes but remains challenging for non-coding RNAs (ncRNAs) due to weak sequence conservation, ambiguous transcript boundaries, and incomplete annotations. Here we present CURIA, a pipeline for cross-species ncRNA annotation that combines genome alignment chains with embedding-based representations derived from the RiNALMo RNA foundation model. CURIA formulates cross-species ncRNA correspondence as synteny-constrained matching in embedding space. It first identifies candidate loci using a simplified TOGA-based projection, then detects localized regions whose embedding patterns resemble those of annotated compact ncRNAs enriched for structured RNA classes (“islands”), and finally establishes cross-species correspondence by local alignment in embedding space. We applied CURIA to 19 human-to-query genome pairs. In the detailed human-to-mouse annotation comparison, 52.4% of predicted short-ncRNA loci overlapped an annotated exon (any overlap). For compact short ncRNAs, embedding similarity was largely redundant with the tested conventional sequence metrics, whereas localized embedding-SW matching of island regions provided additional ranking information beyond nucleotide alignment, identity, k-mer similarity, composition, and length in the tested human–mouse and human–cow samples. From 34,846 annotated human lncRNA loci, a multi-species screen identified a broad recurrent set of 1,693 annotated human lncRNA loci with an embedding-supported region in at least 17 of 19 query genomes; 170 loci remained when every counted species was additionally required to satisfy $d \leq 0.02$. These results suggest that combining synteny with embedding-based similarity can complement conventional sequence conservation for cross-species ncRNA annotation in settings characterized by weak or heterogeneous primary-sequence conservation, establishing a proof of concept for integrating comparative genomics with RNA foundation-model embeddings for genome-scale ncRNA analysis.

1 Introduction

1.1 Protein-coding orthology as a tractable problem

Orthology inference for protein-coding genes is well established and supported by scalable annotation transfer pipelines [1–7]. Strong sequence conservation, together with constraints imposed by open reading frames, enables reliable cross-species mapping of gene loci [2, 4]. Methods such as TOGA, which combine genome alignment chains with coding sequence structure, achieve high accuracy at genome scale. However, these approaches rely on properties specific to protein-coding genes and do not generalize to non-coding RNAs. In particular, they assume conserved coding structure and detectable sequence similarity, which are often absent outside protein-coding contexts.

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In this work, we do not attempt to infer evolutionary orthology. Instead, we focus on identifying candidate cross-species correspondences supported by genomic synteny and consistency in embedding space. This formulation is best viewed as synteny-constrained annotation transfer, where predicted loci represent candidate corresponding elements rather than definitive orthologs.

1.2 The ncRNA orthology challenge

Establishing cross-species correspondence for ncRNAs remains substantially more difficult. Sequence conservation is often weak or absent across evolutionary distances [8–13], and transcript boundaries are frequently ambiguous, making it unclear what constitutes a conserved unit [8–10, 14]. Functional constraints are not reliably captured by primary sequence alone, limiting the applicability of sequence-based approaches [11]. In addition, ncRNA annotations are incomplete and heterogeneous across species [14, 15], complicating both evaluation and transfer. Finally, embedding representations of short ncRNAs can be context-dependent [16], making direct comparison sensitive to boundary definition.

This ambiguity is especially acute for lncRNAs. Their annotated loci can span tens of kilobases, often with localized candidate domains separated by long intervening sequence. Whole-locus comparison can therefore dilute the detectable signal, and the locus may exceed the sequence length supported by current RNA foundation models. This motivates searching for localized candidate cores rather than requiring end-to-end transcript correspondence.

For example, two ncRNA loci may exhibit low primary sequence similarity while preserving a similar structural pattern:

```
GAGGUAUCCAAGAUACCUC
(((((((.....))))))
AUGCUACGAUCGGUAGCAU
(((((((.....))))))
```

In such cases, sequence-based comparison may fail to establish correspondence, whereas embedding-based representations may capture similarities associated with shared structural patterns.

1.3 Limitations of existing approaches

Existing approaches solve related but distinct problems. Covariance models (e.g., Infernal [17]/Rfam [18]) offer high specificity but rely on curated families and do not generalize to genome-wide discovery. Structure-based alignment methods such as LocARNA [19] and RNAforester [20] capture explicit sequence–structure similarity between supplied RNAs. CURIA instead starts from genomes, a reference annotation, and alignment chains, and localizes candidate corresponding elements without requiring a pre-specified family model or structure. Genome-wide methods such as RNAz [21], EvoFold [22], and FEELnc [23] identify candidate RNA elements or lncRNA loci, but focus on detection or annotation rather than establishing cross-species correspondence. Comparative genome alignments define candidate homologous intervals but do not provide an RNA-specific measure of similarity or a principled way to localize relevant embedding-supported signal [24, 25]. The distinguishing objective here is annotation transfer with localized RNA-specific evidence: first identify the relevant syntenic locus, then find and compare subregions with embedding patterns characteristic of annotated compact ncRNAs enriched for structured RNA classes.

1.4 Synteny-constrained embedding matching

An underexplored component is the use of pretrained RNA-language-model representations as a localized similarity signal within syntenically constrained genomic search regions. Genome alignment chains already identify regions that share genomic context and therefore indicate where

a corresponding ncRNA should be sought [24, 25]. At the same time, RNA foundation models such as RNA-FM [16], RiNALMo [26], and related embedding approaches such as RNABERT [27] provide learned representations associated with RNA family identity, secondary-structure-related properties, and downstream predictive tasks [28]. Applying these models as an unconstrained genome-wide scanner would be impractical; using chain-defined regions makes local RNA-specific comparison tractable. Combining these sources of information enables tractable comparison of ncRNA loci in settings where primary-sequence conservation is limited.

1.5 Method overview

We formulate cross-species ncRNA correspondence as matching of embedding-supported regions constrained by genome alignment chains. Candidate loci are first identified using a TOGA-inspired projection that leverages alignment coverage in intronic and flanking sequence to provide locus-level syntenic context that is partly independent of conservation within the ncRNA itself [1, 2].

Within these candidate regions, CURIA evaluates similarity in embedding space rather than relying on primary-sequence alignment. The pipeline treats the short and island branches with complementary similarity measures. For short ncRNAs, each region is represented as a set of per-position embedding vectors and similarity is quantified using Maximum Mean Discrepancy (MMD) [29], a distribution-level measure over embeddings; low MMD indicates similar embedding distributions. In the island branch, localized detector-positive subregions are matched by token-level local alignment using Smith–Waterman alignment [30] over the per-token cosine similarity dotplot. Both formulations avoid explicit primary-sequence alignment and provide continuous scores for ranking candidate matches, making them suitable for ncRNAs where primary-sequence conservation is weak.

1.6 Contributions

In this work, we:

1. Reformulate cross-species ncRNA correspondence as synteny-constrained matching in embedding space.
2. Introduce CURIA, a pipeline combining chain-based locus projection with embedding-based detection and comparison of localized signal.
3. Introduce an island-based representation for geometry-defined island-branch loci that separates detectable candidate cores from intervening sequence not required to align.
4. Develop a boundary refinement strategy for short ncRNAs.
5. Demonstrate a genome-scale proof of concept across multiple mammalian genomes, showing that this approach can recover cross-species ncRNA correspondence in cases where primary-sequence similarity alone is limited.

2 Methods

2.1 Pipeline overview

CURIA is a computational pipeline for genome-scale cross-species ncRNA correspondence that combines chain-guided locus prediction with embedding-based comparison (Figure 2). Given a reference genome with ncRNA annotations and one or more query genomes linked by genome alignment chains, CURIA produces: (i) predicted corresponding loci and (ii) an embedding-based

similarity score for short ncRNAs; for island-branch loci it instead reports (iii) matched localized subregions (“islands”).

The pipeline follows a branching architecture after an initial projection step, handling short and island-branch loci separately. Short ncRNAs yield a single refined locus, whereas island-branch loci yield a set of island-level alignments. The pipeline consists of seven stages:

1. **Preprocessing.** Isoforms of the same gene/locus and strand are merged into a *union exon model* (their exonic intervals unioned in genomic order).
2. **Candidate region prediction.** Genome alignment chains are used to identify candidate query regions via a simplified TOGA-inspired classifier (Section 2.3).
3. **Reference island detection (island branch).** Localized subregions (“islands”) are detected within transcripts (Section 2.7).
4. **Chain-guided island projection (island branch).** Reference islands are projected to the query genome to define targeted search regions (Section 2.8).
5. **Query island detection (island branch).** Candidate islands are identified in projected regions (Section 2.9).
6. **Island alignment (island branch).** Reference and query islands are matched by embedding-based alignment (Section 2.10).
7. **Short ncRNA refinement (short branch).** Whole candidate loci are refined by embedding-based local boundary search (Section 2.5), without island decomposition.

Outputs include candidate region assignments, refined short ncRNA loci, and island-level alignments.

Branch routing. Assignment to the short or the island branch is based on transcript geometry rather than annotation biotype. A locus is routed to the short-ncRNA branch when its union exon model (defined above) is single-exon and ≤ 256 nt; all other loci are routed to the island branch. Within the island branch, single-exon loci ≤ 320 nt are treated as a single whole-exon island (compact-RNA handling; Section 2.7), whereas longer or multi-exon loci are decomposed into islands by the detector. Concretely: single-exon ≤ 256 nt $\rightarrow$ short branch; single-exon 257–320 nt $\rightarrow$ island branch as one whole-exon island; single-exon > 320 nt and all multi-exon loci $\rightarrow$ island branch with island detection. Routing is a partition—no locus is processed by more than one branch. Consequently some loci annotated as lncRNA that are short and single-exonic are evaluated in the short-ncRNA benchmark, while some non-lncRNA classes that are multi-exonic are processed through the island branch.

The union exon model is an exploratory locus-level representation for signal discovery, not a reconstruction of a naturally occurring mature transcript: as it unions exons across isoforms of a locus, mutually exclusive isoforms may be combined. It is used only to define where to search for candidate localized signal, and no biological claim is attached to the merged model itself.

2.2 Input data

CURIA requires three types of input data:

1. **Reference genome and annotation.** The reference genome is provided in 2bit format, with ncRNA annotations in BED12 format including `transcript_id`, `gene_id`, and `biotype`. Protein-coding transcripts are used during candidate region prediction to compute syntenic-related features, but are excluded from downstream analysis.
2. **Query genomes.** Query genomes are provided in 2bit format. Query-side annotations are not required for prediction but may be used for evaluation.

3. **Genome alignment chains.** Pairwise genome alignment chains (UCSC chain format) define syntenic relationships between reference and query genomes. Low-scoring chains are filtered during loading.

2.3 Candidate region prediction

Input. Union transcripts and genome alignment chains.

Operation. CURIA uses a simplified TOGA-inspired chain-guided projection step [1] to define coarse candidate regions for ncRNA loci rather than full transcript reconstructions (Figure 2A). For each chain–transcript pair, a small set of seven chain- and exon-structure features is computed: log-transformed syntenic gene count (`synteny_log1p`), global exonic occupancy (`gl_exo`), ± 10 kb flank coverage (`flank_cov`), exonic-length to query-span ratio (`exlen_to_qlen`), exonic fraction (`exon_perc`), exon count (`ex_num`), and a single-exon indicator (`single_exon`).

A gradient-boosted decision-tree classifier assigns a correspondence probability from these features, trained on labeled chain–transcript pairs from human–mouse projections. Pairs with predicted probability above 0.5 are retained, except that the deployed model uses 0.3 for loci annotated as lncRNA to favor candidate-region recall. Because it uses only chain- and exon-structure features, this step is independent of the choice of RNA embedding model. Optionally, for deeply diverged query genomes where the orthology classifier is over-conservative, the search space can instead be seeded from the top-scoring alignment chain per transcript regardless of the orthology call.

For each retained pair, candidate query intervals were obtained by projecting the transcript through the corresponding alignment chain. Intervals were defined as the span of chain blocks overlapping the transcript, including flanking regions to account for alignment imprecision. When multiple chains satisfied the threshold for a given transcript, all corresponding intervals were retained.

Output. Projected candidate query regions with associated metadata.

Rationale. Only coarse locus-level projections are required to constrain downstream analysis, avoiding assumptions related to coding sequence structure and transcript reconstruction.

2.3.1 Classifier training and evaluation partitioning

Chain–transcript pairs are labeled by the orthology procedure the step adapts [1]: pairs classified as orthologous (ORTH) form the positive class and pairs classified as paralogous (PARA) the negative class; other categories are excluded from training. Each of the seven features is derived only from chain geometry and reference-side exon structure (syntenic coverage, gene-level exonic overlap, flank coverage, exon-to-query length ratio, exon fraction, exon count, and a single-exon flag); no RNA foundation-model embedding and no query-species ncRNA annotation enter the labeling or the features. A gradient-boosted decision-tree classifier is trained on a balanced sample of up to 20,000 pairs per class with a random 75/25 stratified train/test split (fixed seed), and generalization is summarized by five-fold stratified cross-validated AUC. The split is stratified by label but *not* grouped by transcript, gene, locus, chromosome, or family, so paralogous or neighboring loci can in principle fall on both sides of the split; the reported classifier AUC should therefore be read as an estimate of ORTH-versus-PARA feature separability rather than a strict held-out-locus generalization bound. Crucially, because the features are embedding-independent and the classifier only gates the candidate search space, it is independent of the embedding-based matching that the downstream annotation-agreement analysis evaluates; that evaluation compares predicted loci to query annotations the classifier never sees, so classifier training does not directly confound it.

2.4 RNA embeddings

Sequences are represented by per-nucleotide embeddings from the RiNALMo general-purpose RNA foundation model (**giga-v1**; 650M parameters, 1280-dimensional per-token embeddings) [26].

Two task-specific linear projections are stored as pipeline artifacts: a $k=16$ PCA used for position-level matching (fit on 10,368 per-token embeddings), and a $k=64$ PCA used for island finding (fit on the 18,827 mean-pooled training windows described below). Per-model parameters (PCA files, signal/noise classifier, scan and match thresholds, embedding strategy) are recorded in a model registry and selected automatically.

2.5 Short ncRNA pipeline

Short ncRNAs (e.g., miRNAs, tRNAs, snoRNAs, snRNAs) typically occupy compact loci, allowing direct local comparison without island-based decomposition (Figure 2C). This branch processes single-exon ncRNAs up to 256 nt. For each locus, the reference sequence is embedded and the corresponding query region is extended symmetrically toward a search span 1.1 times the reference length when additional sequence is needed. Candidate windows matching the reference length are tiled across the query region and compared to the reference using MMD (Section 2.6). The best match is then refined by local boundary perturbation (± 5 nt), constrained to a length within $0.8\text{--}1.2\times$ the reference length. Embeddings are projected from 1280 to 16 dimensions using the matching PCA prior to comparison.

Output. Refined query coordinates with an MMD-based similarity score.

2.6 Similarity in embedding space (MMD)

For short ncRNAs, window similarity is quantified using Maximum Mean Discrepancy (MMD) [29] with an RBF kernel, treating per-position embeddings as empirical distributions:

$$\widehat{\text{MMD}}_u^2(X, Y) = \frac{1}{n(n-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j), \quad (1)$$

i.e. the unbiased empirical estimator (self-terms $i = j$ excluded from the within-sample sums), where $k(x, y) = \exp(-\gamma\|x - y\|^2)$ and $X = \{x_1, \dots, x_n\}$, $Y = \{y_1, \dots, y_m\}$ denote the sets of per-position embedding vectors for two sequence windows. The implementation reports $\sqrt{\max(\widehat{\text{MMD}}_u^2, 0)}$, because the unbiased finite-sample estimate can be negative. Thus a reported value of zero includes estimates clipped at zero and does not uniquely imply numerically identical empirical distributions. Low MMD indicates similar embedding distributions; higher values correspond to dissimilar or background signal.

The kernel bandwidth γ is set by the median pairwise-distance heuristic computed on the reference locus’s own per-position embeddings and then held fixed for all query-window comparisons at that locus. Consequently γ is chosen per reference locus rather than globally. The score is directly comparable for ranking windows *within* a locus, but intermediate absolute values (e.g. 0.05 vs 0.1) are not guaranteed to be on a single globally calibrated scale across loci or biotypes. Cross-locus comparisons and global thresholds are therefore interpreted as approximate rather than exact.

2.7 Reference island detection

To detect localized embedding patterns resembling annotated compact ncRNAs enriched for structured RNA classes, we trained a logistic regression classifier on fixed-length sequence windows represented in RiNALMo embedding space. Positive examples consisted of windows from annotated miRNAs, snoRNAs, snRNAs, scaRNAs, and misc_RNAs (lncRNAs were excluded),

labeled positive when a window covered ≥ 20 nt of an annotated element; negative examples consisted of intergenic, non-exonic genomic windows. Windows were embedded with RiNALMo, mean-pooled across positions, and projected from 1280 to 64 dimensions using the finding PCA prior to training. Training loci were balanced across biotypes (up to 150 loci per biotype, with an additional 71 held-out loci per biotype for calibration). The committed training cache contains 18,827 windows: 6,538 positive and 12,289 negative. On 5-fold cross-validation the classifier achieved ROC-AUC ≈ 0.79 ; at a calibrated background false-positive budget of $\leq 20\%$, it detected $\approx 78\%$ of signal windows (≥ 40 nt overlap).

The resulting model distinguishes embedding patterns associated with the annotated compact ncRNAs in its positive set from genomic background. We call contiguous detector-positive regions “islands.” This is an operational embedding-based definition, not evidence that every island forms a stable structure or has a biological function, and the model is not a general ncRNA annotation classifier.

Each reference union transcript is scanned using a sliding window (128 nt, stride 40 nt). Windows are embedded, mean-pooled, projected to the finding PCA space, and scored by the classifier. The resulting probability trace is smoothed using a moving-average filter (window size = 5), and contiguous regions exceeding a fixed probability threshold (0.5) are defined as candidate islands. Island coordinates are mapped to genomic space, allowing multi-segment islands across splice junctions. Compact single-exon transcripts (exon ≤ 320 nt, e.g. RMRP) are not scanned but taken as a single island spanning the whole exon. This is an engineering heuristic that avoids unstable decomposition of compact loci; it does not assume end-to-end biological conservation.

Output. Reference islands with genomic coordinates and associated scores.

2.8 Chain-guided projection of islands

Input. Reference islands, genome alignment chains, and candidate region assignments.

Reference islands are projected through alignment chains to define candidate query regions (Figure 2A). At most five chains per gene are retained (lower chain ID denotes higher alignment score), and chains are loaded with a minimum score of 25,000. Before projection, each reference island is extended by 144 nt on each side. Projections exceeding 25 times the flanked reference length are rejected. Each accepted query projection is then extended on each side by the unflanked island length plus 256 nt. Projected intervals on the same chromosome and strand are merged when they overlap, while retaining links to the originating reference islands.

Output. Query regions with mappings to reference islands.

2.9 Local query island evaluation

Input. Merged query regions, query genome, and the classifier from Section 2.7.

Only regions linked to a transcript with at least one reference island are scanned. Regions shorter than 128 bp or longer than 1.5 Mb are excluded. For regions longer than 2.5 kb, at least one linked reference transcript must be nonzero in length and no more than fourfold shorter than the query region. Each retained query region is scanned using the same window, stride, smoothing, and probability threshold as reference island detection; contiguous high-scoring windows define query islands.

Output. Query islands with genomic coordinates and scores.

2.10 Island alignment in embedding space

Each island (reference and query) is embedded as a complete sequence, and its per-token embeddings are projected to the matching PCA space ($1280 \rightarrow 16$). For a reference island with per-token matrix $A \in \mathbb{R}^{L_r \times 16}$ and a query island $B \in \mathbb{R}^{L_q \times 16}$, we L2-normalize the rows

and form the per-token cosine similarity dotplot $C = \hat{A}\hat{B}^\top$, where C_{ij} is the cosine similarity between reference position i and query position j .

Nucleotide-indexed local alignment paths are extracted by a Smith–Waterman procedure [30] on the score matrix $S = C - \tau$, with cosine offset $\tau = 0.5$ and a linear gap penalty of 0.3. From the embedding Smith–Waterman alignment score s (which integrates cosine similarity over the aligned band length) we define the embedding-SW distance $d = 1/(1 + s)$; lower d indicates a stronger match. Candidate matches are filtered by minimum island length (≥ 72 nt), minimum effective aligned length (≥ 40 nt), and match distance ($d \leq 0.1$). Within each projected locus, matches are then assigned greedily in order of increasing d : each query island is claimed by at most one reference island, and each reference island accepts at most two query islands (so a reference island with a duplicated query copy can retain both, while further copies are left unassigned). Exact distance ties retain the deterministic reference-index, query-index order used to enumerate candidate pairs. A collinearity constraint retains a maximum-cardinality non-decreasing sequence of query-island indices on the same strand, or a non-increasing sequence on opposite strands; within a reference island, matched subregions must also be non-decreasing in reference position. Assignment is performed independently per locus rather than globally, and the retained matches with their distances d are reported.

Because s accumulates cosine similarity along the alignment path, $d = 1/(1 + s)$ decreases with the aligned length as well as with per-position similarity: a long alignment of moderate mean cosine can attain a smaller d than a short alignment of high mean cosine. The reported distance therefore combines match length and match quality, which should be kept in mind when thresholding ($d \leq 0.1$), ranking, greedy assignment, and comparing d across RNA classes or across species of differing divergence (Figure 6C). We report d as an empirical, length-dependent match score rather than a length-normalized similarity, and additionally record the effective aligned length for each match. Across the panel, d is moderately anticorrelated with effective aligned length (Spearman $\rho \approx -0.64$; median matched length ≈ 150 nt), and median d is lower for longer alignments (≈ 0.03) than for shorter ones (≈ 0.06), confirming that the score conflates length and per-position similarity; we therefore treat d as a within-locus ranking statistic rather than a calibrated cross-locus similarity, and do not length-normalize.

In the Rfam-derived matcher benchmark (`notebooks/matching.benchmark.ipynb`), positives were pairs sharing a curated Rfam seed-family core embedded in unrelated genomic flanks, whereas negatives contained different-family cores. The embedding-SW alignment score separated these planted match and non-match pairs with AUC 0.993 and localized the planted core in all 140 positive pairs. This is a matcher sanity check under controlled flanks, not an estimate of genome-wide annotation accuracy.

We additionally evaluated matched islands in the human–cow (hg38–bosTau9) and human–mouse (hg38–mm39) comparisons using two null analyses. First, 250 matched island pairs per comparison were sampled at evenly spaced ranks from the complete match table. After embedding each reference island, its assigned query island in both orientations, and a dinucleotide-preserving shuffle of that query sequence, we applied the same cosine-dotplot Smith–Waterman scorer used by the pipeline. We ranked the assigned query among all 250 query islands, calculated an AUC for assigned versus cross-locus pairs, and compared assigned-pair distances with the shuffle control. Second, for 400 matched islands per comparison, we tiled same-length windows across the projected query locus at a stride of half the assigned-island length (minimum 25 nt), retaining at most 100 evenly spaced windows and always including the assigned interval. Windows whose midpoints lay within one assigned-island length were excluded as overlapping alternatives. For loci with at least one remaining alternative, we recorded whether the assigned interval outscored every alternative; its locus-specific chance probability under exchangeability is $1/(k + 1)$ for k alternatives. These analyses test cross-locus specificity, sensitivity to local nucleotide composition, and within-locus positional specificity; they do not establish biological function.

In a separate sequence-baseline analysis, we tested whether embedding similarity was fully

reducible to conventional nucleotide similarity by evaluating the short and island branches separately against the same panel of sequence-derived metrics: normalized Levenshtein identity, normalized affine-gap nucleotide-SW score, aligned-region identity, 3-mer and 4-mer cosine similarity, dinucleotide-composition distance, length ratio, and GC-content difference. For 2,994 human–mouse short-ncRNA predictions, we compared gene-grouped, five-fold cross-validated logistic models for any-exon annotation support, using either the sequence metrics alone or the same metrics plus MMD. For island matching, 200 CURIA-assigned partners were sampled separately from the human–mouse and human–cow comparisons and contrasted with cross-locus alternatives; models were evaluated with reference-island-grouped cross-validation and cluster-bootstrap confidence intervals. A prespecified low-sequence-similarity subset comprised the lowest quartile of normalized nucleotide-SW among assigned pairs. Within each syntenic locus, we also compared the rank of the CURIA-assigned partner with same-length tiled alternatives. All island AUC calculations used freshly recomputed embedding-SW scores for both assigned and alternative pairs; the stored match table is threshold-selected at $d \leq 0.1$ and therefore has a restricted score range.

2.11 Evaluation metrics

Evaluation is based on agreement with existing annotations and serves as an approximate proxy for ground truth. Due to incomplete and heterogeneous ncRNA annotation across species, these metrics should be interpreted as measures of annotation agreement rather than direct estimates of biological accuracy.

2.11.1 Short ncRNA evaluation

For short ncRNAs, evaluation is performed at the locus level against annotated transcripts using three explicitly distinguished criteria:

- *any-overlap rate*: the fraction of predictions intersecting an annotated ncRNA exon by at least 1 nt;
- *50% prediction-coverage recovery*: the fraction for which the maximum exonic intersection with an annotated transcript, divided by the predicted-locus length, is at least 50%;
- *near-complete boundary recovery*: the fraction satisfying a same one-directional prediction-coverage criterion at 99%.

These labels are used consistently throughout; the ambiguous term “annotation agreement” is avoided for the three distinct quantities.

We separately report workflow coverage from the full reference denominator: the number of non-protein-coding union loci satisfying the short-branch geometry, the number assigned at least one chain-projected query candidate, and the number producing a refined prediction. This is a processing-yield denominator, not a biological recall estimate. A recall denominator would require a linked set of eligible human loci with established mouse orthologs; query-side interval overlap alone does not establish that relationship. As a conservative proxy, we also define a *named-counterpart* set: eligible human loci whose gene name occurs exactly, ignoring case, in the mouse GENCODE annotation. Recovery requires a final prediction to overlap an exon of that same-name mouse gene. This criterion favors established, consistently named RNA families and is not a general genome-wide recall estimate.

2.11.2 lncRNA locus-level evaluation

For lncRNAs, evaluation is performed at the locus level with relaxed criteria due to boundary ambiguity. A predicted locus is considered supported if it overlaps any annotated exon of an

ncRNA gene. We report the fraction of loci overlapping annotated exons and the distribution of overlap proportions across predictions.

2.11.3 Island-level evaluation

For island-branch loci, evaluation is extended to localized subregions (“islands”). A matched island is considered supported if it overlaps any annotated ncRNA exon in the query genome. We report the fraction of matched islands overlapping annotated regions, the number of matched islands per locus, and cross-species consistency of island matches.

2.11.4 Annotation agreement versus confidence

To assess the relationship between embedding-based similarity and annotation support, we analyze annotation agreement as a function of the match score. Predictions are stratified by score and we report the fraction overlapping annotations across bins and the score distributions for overlapping and non-overlapping predictions. This provides an empirical characterization of the score as a measure of correspondence confidence, rather than relying on predefined thresholds.

2.11.5 Limitations

Predictions outside annotated regions may represent either false positives or unannotated ncRNAs [10, 14, 15]. Annotation completeness varies across species, and ncRNA boundaries are often imprecise. In contrast to protein-coding ORFs, many ncRNAs have no comparably sharp, independently verifiable functional boundary. Accordingly, overlap-based performance is a conservative proxy rather than a precise boundary benchmark.

2.12 Recurrent-core aggregation and coding proximity

Reference islands from different species runs are assigned to the same core when their human reference intervals overlap within the same gene; the best-distance match per core and query species is retained. A species contributes to a core when it contains an accepted island match; accepted rows already satisfy the pipeline threshold of embedding-SW distance $d \leq 0.10$. The broad recurrent set requires an accepted match in at least 17 of 19 query species and mean reference-island length of at least 120 bp. Because every retained per-species row is individually bounded by $d \leq 0.10$, applying the same cutoff to the cross-species mean adds little further discrimination. In the strict-quorum sensitivity analysis, we instead counted species whose individual best match satisfied $d \leq 0.05$, 0.04, 0.03, 0.02, or 0.01 while fixing the recurrence requirement at 17 species; we also varied the species quorum at $d \leq 0.02$. Human lncRNA gene spans are then compared with protein-coding gene spans. Direct overlaps are classified by strand as sense, antisense, or both; non-overlapping loci within 5 kb are classified as near coding; all remaining loci are classified as intergenic. This is a coding-proximity description of annotated human lncRNA loci, not a test of transcription or function.

2.13 Exploratory ViennaRNA structure comparison

The exact human–mouse intervals for the three short-ncRNA examples in Figure 5 and the accepted MALAT1 recurrent-core matches R5, R4, R3, and R2 were analyzed with ViennaRNA 2.7.2 [31]. The similar-length short-ncRNA intervals were folded whole. For each MALAT1 core, length-matched windows were defined from the aligned span and expanded symmetrically by 20 or 40 nt. Global partition-function folding and RNAplfold were applied, and the human and mouse sequences were aligned by affine-gap nucleotide Smith–Waterman. Both base-pair probability matrices were mapped into the shared alignment coordinates and compared using

column-exact weighted Jaccard and a ± 2 -column tolerance-aware weighted overlap with one-to-one matching. These structural-overlap metrics are exploratory descriptive measures, not validated structure-similarity scores. Full per-window and per-method results are provided in Supplementary Figures S1–S7, Supplementary Table S1, and the source TSVs.

2.14 Implementation details

Embeddings are computed with RiNALMo and projected using the two task-specific PCA transformations described in Section 2.4 ($1280 \rightarrow 16$ for matching and $1280 \rightarrow 64$ for island finding). Short ncRNA loci are embedded per-window, whereas complete island embeddings supply the per-token matrices used by the local matcher.

Embedding inference is performed on GPU using batched execution, whereas downstream MMD computation, cosine alignment, and filtering are performed on CPU. Internally, CURIA is organized as a staged task pipeline that separates embedding inference from downstream analysis: sequence windows are processed by a shared GPU worker using micro-batching, while subsequent matching and filtering steps operate on persisted intermediate results. This design improves hardware utilization, supports incremental execution, and allows reference-side preprocessing to be reused across query genomes. Attention uses PyTorch’s memory-efficient scaled-dot-product attention. Outputs are reported in standard tabular and BED formats. The use of fixed embedding models and PCA projections ensures consistent representations across runs.

3 Results

3.1 Benchmark setup

CURIA was evaluated on 19 human-to-query genome pairs, using human (hg38) as the reference, spanning primates through the marsupial outgroups: rhesus macaque (rheMac10), mouse (mm39), rat (rn7), rabbit (HLoryCun3), cow (bosTau9), pig (susScr11), Bryde’s whale (HLbalEde1), dromedary (HLcamDro2), horse (equCab3), cat (felCat9), dog (canFam5), pangolin (HLmanJav2), flying fox (HLpteVam2), hedgehog (eriEur2), armadillo (dasNov3), Asian elephant (HLeleMax1), and three marsupials: gray short-tailed opossum (monDom5), Virginia opossum (HLdidVir1), and tammar wallaby (HLnotEug3). Genome assemblies were obtained from NCBI [32] and the UCSC Genome Browser [33]. Due to variability in annotation quality across species, high-quality ncRNA annotations were used only for human (GENCODE V48) and mouse (GENCODE VM37) [34], while other species were analyzed without relying on their native annotations. Human annotation contains thousands of lncRNA loci, but biological evidence and mechanistic characterization are concentrated in a much smaller subset; the remainder comprises a mixture of functional transcripts, transcription-associated products, and uncertain annotations [13, 14]. Evaluation was performed at both the locus level (for short ncRNAs and lncRNA loci) and the substructure level (for island matches), following the criteria defined in Section 2.11.

3.2 Overall performance

We first quantified short-branch processing yield and agreement between predicted loci and existing annotations. Of 9,247 annotated non-protein-coding human union loci satisfying the single-exon, ≤ 256 -nt short-branch geometry, 2,949 were assigned at least one chain-projected mouse candidate locus and 2,948 produced a refined prediction. Multiple retained chains produced 2,994 prediction rows, which form the denominator for the following query-annotation agreement analysis; these counts measure workflow coverage, not recall against known orthologs. Among the 2,994 predictions, 52.4% show any overlap with an annotated exon (≥ 1 nt; any-overlap rate), 46.5% have $\geq 50\%$ of the predicted-locus length covered by an annotated exon, and 26.0% have $\geq 99\%$ prediction coverage. These are one-directional prediction-coverage criteria, not reciprocal

overlap. Among any-overlap predictions, the median overlap is 98.9% (mean 86.8%), indicating that refined predicted loci are typically well localized and consistent with annotated boundaries.

Under the conservative exact-name criterion, 446 geometry-eligible human loci had an annotated mouse gene with the same name; CURIA predictions overlapped an exon of that same-name mouse gene for 373 loci (83.6% rough named-counterpart recall). Because this subset is enriched for established, consistently named RNA families, it should not be interpreted as recall over all short ncRNAs.

Refined predicted loci are generally well localized when annotation support is present.

3.3 Short ncRNA performance and score behavior

We evaluated similarity between matched short ncRNA loci using MMD in embedding space. For the human–mouse pair (2,994 loci), the MMD distribution is skewed toward low values (median 0.142; Fig. 4A), with a substantial fraction of near-zero values (32.5% below 0.05), indicating consistent embedding representations between matched reference and query loci.

Stratification by biotype reveals clear differences (Fig. 4C). snoRNAs and tRNAs exhibit the lowest MMD values (median ≈ 0), consistent with their strong, well-characterized structural conservation [35]. miRNAs and lncRNAs occupy an intermediate range, whereas snRNAs, misc_RNAs, and rRNA-derived loci show higher and broader distributions. The rRNA category is too small and heterogeneous here for a biological conclusion about rRNA conservation; its distribution may instead reflect copy assignment, pseudogene contamination, projection, or boundary effects.

MMD is strongly associated with annotation overlap. Predictions with low MMD values ($\lesssim 0.1$) show high any-overlap rates with annotated loci ($\sim 88\%$), whereas higher MMD values show reduced overlap and increased dispersion. Predictions with $\text{MMD} > 0.2$ rarely overlap annotations and are enriched for ambiguous cases. Consistent with this, loci overlapping annotations are enriched at low MMD values, while non-overlapping predictions are shifted toward higher values (Fig. 4B), supporting the interpretation of MMD as a continuous empirical association with annotation support in this benchmark, rather than a globally calibrated confidence scale.

A subset of predictions with very low reported MMD does not overlap existing annotations. Zero is the minimum reported score but can result from clipping a negative unbiased estimate.

Embedding similarity was strongly associated with nucleotide similarity but was not fully reducible to the tested sequence-derived metrics. Across all 2,994 short-ncRNA predictions, MMD was anticorrelated with nucleotide identity (Pearson $r = -0.80$; Spearman $\rho = -0.85$). A multivariate sequence-only model predicted annotation support with ROC-AUC 0.887, compared with 0.893 after adding MMD ($\Delta\text{ROC-AUC} = +0.0064$, 95% CI [0.0030, 0.0097]). This small increment disappeared when miRNAs were excluded ($\Delta = -0.0007$), indicating that MMD adds little predictive information over the full sequence baseline for the remaining short-ncRNA classes in this benchmark.

3.4 Localized embedding-supported regions (“islands”)

We analyzed localized subregions (“islands”) within lncRNAs to assess cross-species correspondence beyond the locus level. Islands are matched at token level by Smith–Waterman local alignment over the per-token cosine similarity dotplot, and match quality is reported as an embedding-SW distance $d = 1/(1 + \text{score})$ (lower is better; Section 2.10). Island detection reveals that detector-positive embedding patterns are sparse and localized. Across the 19 comparisons, 34,532 of 34,846 annotated human lncRNA union loci entered reference-island processing in at least one comparison, and 18,364 (53.2%) had at least one reference island. Among island-positive loci, the mean was 1.51 reference islands and the median was 1 (27,784 detector islands in total), indicating a small number of localized regions rather than signal distributed uniformly across the

transcript (Fig. 6A). Because an island can cross splice junctions, the 27,784 islands comprise 43,506 genomic segments.

To assess cross-species correspondence, we evaluated matches between reference and query islands. About half of matched islands (50.1%) overlap annotated ncRNA regions in the query genome, and 28.6% show $\geq 50\%$ exonic overlap.

Together, these observations support a modular pattern of cross-species correspondence, with signal concentrated in discrete localized regions rather than uniformly distributed across entire transcripts [10–12, 14, 36]. This interpretation is broadly consistent with observations from well-characterized lncRNAs such as NEAT1, XIST, and MALAT1, in which localized domains have been linked to distinct architectural, spreading, or stability-related functions [37–39].

3.4.1 Multi-species recurrent cores

Here, “core” refers operationally to an island showing recurrent accepted matches across multiple species within syntenically projected loci. We observe a subset of islands that exhibit reproducible cross-species matches (Fig. 6B–C). Figure 6B starts from 31,977 reference-coordinate cores having at least one accepted match in any query genome; it therefore excludes loci without a detected reference island and islands with no accepted match. Within this conditional denominator, 87% are recovered in at least two query genomes, 67% in five or more query genomes, and approximately 2% are detected across all 19 query genomes, including the marsupial outgroups (opossums and wallaby). An absent species in this aggregation can reflect either missing projection or failure to obtain an accepted match; the bar plot does not separate those outcomes. The two counts use different universes. The 27,784 detector islands are unique islands detected in human lncRNA loci before projection. The 31,977 cores are reconstructed from accepted match rows across all 19 species runs and all island-branch biotypes; species-run-specific genomic segments are subsequently aggregated by overlap in human reference coordinates.

In contrast to locus-level overlap, which is often incomplete or ambiguous, these cores exhibit recurrent cross-species correspondence at the level of localized embedding similarity. To assess whether such matches could arise from selecting the best of many candidate positions within a syntenic window, we compared each matched query island against same-length windows tiled across its own projected syntenic locus (a within-locus positional null; representative genome pairs). The assigned match showed strong cross-locus specificity: among 250 pairs per comparison, the assigned query ranked in the top 5% for 69% of human–cow and 65% of human–mouse reference islands, and assigned-versus-cross-locus AUCs were 0.907 and 0.898, respectively. Assigned pairs were closer than their dinucleotide-preserving query shuffles in 94% and 92% of cases. In the within-locus analysis, 287 cow and 308 mouse loci contained at least one non-overlapping alternative. The assigned interval outscored every such alternative in 189/287 (65.9%) and 202/308 (65.6%) loci, compared with mean locus-specific chance probabilities of 15.6% and 15.9%. This enrichment remained well above chance in strata containing 20 or more alternatives (50% versus approximately 3% for cow and 58% versus approximately 3% for mouse). These controls provide evidence against cross-locus, composition-only, and simple best-of-many positional artifacts in the tested loci, without establishing that every matched island is functional. Although substantially above the locus-specific chance expectation, the assigned interval was not the top-scoring non-overlapping alternative in approximately one third of evaluable loci, indicating substantial residual positional ambiguity.

A complementary score-reducibility analysis asked whether the island score adds ranking information beyond nucleotide identity, local sequence alignment, k-mer similarity, composition, and length. Adding embedding-SW to the multivariate sequence-only baseline improved discrimination of CURIA-assigned partners from cross-locus alternatives in both tested samples: ROC-AUC increased from 0.897 to 0.935 for human–mouse ($\Delta = +0.038$, 95% cluster-bootstrap CI [0.022, 0.056]) and from 0.912 to 0.952 for human–cow ($\Delta = +0.040$, [0.026, 0.055]). In the prespecified low-sequence-similarity subset, embedding-SW retained substantially stronger

Table 1: Funnel from annotated human lncRNA gene loci to the near-panel recurrent-core set.

Stage	Unit	Count
Annotated human lncRNA union loci	gene locus	34,846
Entered island processing in at least one comparison	gene locus	34,532
At least one detector island	gene locus	18,364
Projectable in at least 17 query genomes	gene locus	11,328
Gene locus with ≥ 1 core matched in ≥ 17 genomes	gene locus	1,761
After mean reference-island length ≥ 120 bp	gene locus	1,693
Intergenic relative to protein-coding genes (> 5 kb)	gene locus	379

discrimination than nucleotide Smith–Waterman (AUC 0.798 versus 0.615 for human–mouse and 0.796 versus 0.616 for human–cow). Within the same syntenic locus, embedding-SW also ranked the pipeline-selected match first more often than nucleotide Smith–Waterman (31.0% versus 18.4% for human–mouse; 24.4% versus 11.6% for human–cow). This assignment-reproduction analysis evaluates recovery of the CURIA-assigned partner, not whether that partner is the true biological ortholog.

3.5 Near-panel-wide recurrent cores and coding proximity

The 1,693-locus result follows a defined funnel. The reference contains 34,846 annotated human lncRNA union gene loci; 34,532 entered the island branch in at least one genome comparison, 18,364 had at least one reference island, and 11,328 of those were projectable in at least 17 query genomes. Requiring an accepted match for the same reference-coordinate core in ≥ 17 genomes retained 1,761 annotated lncRNA gene loci before reference-length filtering and 1,693 after requiring a mean reference-island length of at least 120 bp. The historical aggregate mean-distance cutoff of 0.10 adds no independent stringency because the individual accepted rows are already bounded at $d \leq 0.10$. The final stages are aggregated by versionless ENSG identifier; thus, 1,693 counts gene loci, not individual islands or de novo lncRNA annotations. This operational threshold allows two biological or technical non-detections; failure to recover a core is not by itself interpreted as lineage-specific absence. Of these loci, 379 are intergenic relative to human protein-coding annotations (no overlap and more than 5 kb from a coding gene). The full set of loci (all conservation categories) is provided as a supplementary table in the repository: https://github.com/kirilenkobm/CURIA_pipeline/blob/master/paper/lncRNAs_with_conserved_cores.tsv. These loci represent candidate recurrent cores within known human lncRNA loci and may include false positives despite filtering. The intergenic subset shows that the screen is not restricted to direct overlap with protein-coding genes, although conservation of other DNA-level regulatory or repetitive elements remains a possible confounder.

Because the matcher accepts island matches at embedding-SW distance $d \leq 0.10$, the broad threshold primarily defines recurrence and coverage rather than stringent per-species match quality. We therefore repeated the 17-of-19 quorum analysis while imposing the same stricter distance cutoff in every counted species. The broad set contained 1,693 loci; requiring $d \leq 0.05$, 0.03, and 0.02 retained 784, 379, and 170 loci, respectively, including 187, 103, and 42 intergenic loci. Even the ultra-stringent $d \leq 0.01$ cutoff retained 21 loci. These nested subsets show that the recurrent signal is not confined to matches near the acceptance ceiling, while also making clear that the broad set is substantially larger than the high-confidence subsets.

Core identity is anchored to the human reference coordinates: overlapping reference intervals are merged across species, whereas non-overlapping intervals within the same gene remain separate cores. Among the cores passing the 17/19 screen, none contained pairwise-disjoint reference intervals joined by transitive overlap.

Table 2: Sensitivity of recurrent-core counts to the per-species embedding-SW distance cutoff at a fixed quorum of at least 17 of 19 query assemblies.

Embedding-SW cutoff	Required species	Cores	lncRNA loci	% broad	Intergenic loci
$d \leq 0.10$	≥ 17	1,813	1,693	100.0	379
$d \leq 0.05$	≥ 17	837	784	46.3	187
$d \leq 0.03$	≥ 17	404	379	22.4	103
$d \leq 0.02$	≥ 17	173	170	10.0	42
$d \leq 0.01$	≥ 17	21	21	1.2	6

3.6 Case studies

To illustrate the behavior of CURIA, we examined individual loci (Fig. 5). Projection of human SNORD57 to mouse identifies a match precisely localized over annotated Snord57, illustrating recovery of a strongly conserved compact ncRNA. Projection of human RNU6-7 intersects an annotated mouse locus (Gm24859). Projection of human vault RNA (ENSG00000199990) identifies a corresponding locus overlapping mouse Vaultrc5; the matched region differs in length from the human transcript, yet CURIA localizes the most similar segment within the syntenic region. Quantitatively, the SNORD57 projection overlaps annotated mouse Snord57 at 74% edit-derived identity (reported MMD 0.00), and RNU6-7 overlaps Gm24859 at 93% edit-derived identity (reported MMD 0.00). The vault-RNA projection overlaps Vaultrc5 despite lower sequence similarity (59% edit-derived identity; normalized nucleotide-SW score 0.35; MMD 0.14), illustrating localization across examples spanning a broad range of sequence conservation.

3.7 Exploratory comparison with predicted secondary structure

We compared the representative short-ncRNA intervals and four accepted MALAT1 recurrent-core matches with alignment-aware ViennaRNA predictions. RNU6-7, used here as a positive control, showed near-identical predicted pairing after mapping the human and mouse base-pair probability matrices into shared alignment coordinates, whereas SNORD57 showed partial agreement. Vault RNA exhibited prominent long-range pairing in both predicted folds but weak alignment-aware contact correspondence. Among the MALAT1 examples, R5 showed the strongest and most stable moderate agreement across the tested boundaries and folding procedures; R4 was weak and boundary-dependent, R3 was weak and inconsistent between global and local folding, and R2 was weak and near background (Supplementary Figures S1–S7 and Supplementary Table S1). These heterogeneous outcomes indicate that embedding similarity is neither interchangeable with nucleotide similarity nor uniformly reflected in ViennaRNA-predicted structural similarity. This exploratory analysis does not establish structural conservation, RNA function, or orthology.

3.8 Error analysis

The dominant failure modes arise from limitations in projection, annotation, and signal strength. Chain-based projection defines the search space and acts as a hard constraint: if the corresponding region is not captured by genome alignments, it cannot be recovered. Assembly gaps, base-level errors, and fragmentation can remove or distort chain coverage and therefore propagate into missed or fragmented predictions. Incomplete and imprecise reference annotations further limit recoverable signal, particularly for lncRNAs. A substantial fraction of loci exhibit weak or diffuse embedding signal, producing no detectable islands or only fragmented matches. Overall, most observed failures are associated with missing or ambiguous signal and projection constraints, rather than instability of the downstream matching procedure.

3.9 Runtime and scalability

We evaluated CURIA on the 19 genome-pair panel. On a single NVIDIA RTX PRO 4500 GPU, a full genome-pair run takes approximately three hours, with runtime scaling approximately linearly with annotation size. Embedding inference is performed on GPU with batched execution, while downstream matching and alignment are CPU-bound; both reference and query island detection are restricted to syntenically projected regions, substantially reducing the effective search space.

4 Discussion

CURIA should be viewed primarily as a proof of concept demonstrating that synteny-constrained embedding comparison yields interpretable cross-species signal for ncRNA correspondence at genome scale, rather than as a finished annotation framework optimized for exhaustive sensitivity or production use. Within this framing, several key observations emerge.

4.1 Key findings

We show that cross-species ncRNA correspondence can be recovered at genome scale using a combination of syntenic constraints and embedding-based similarity. This enables identification of candidate corresponding loci even in the absence of strong sequence conservation. For compact short ncRNAs, MMD provides a useful empirical refinement and ranking score, but in this benchmark it added little predictive information beyond the tested sequence metrics, and the increment disappeared after miRNAs were excluded. The more distinctive result arose in the island branch: localized embedding-SW added ranking information beyond nucleotide alignment, identity, k-mer similarity, composition, and length in the tested human–mouse and human–cow samples. Island-branch correspondence is frequently partial and localized, with signal concentrated in discrete subregions rather than spanning full transcripts. Together, these results support a model in which cross-species ncRNA correspondence is heterogeneous and modular, and can be captured using localized comparison in embedding space.

4.2 Comparison to existing methods

Existing approaches provide complementary capabilities for ncRNA detection and structural comparison but do not directly address genome-scale correspondence. Covariance models require curated family models, while explicit structure alignment compares supplied RNA sequences rather than locating corresponding elements from genome and reference annotation alone. Genome-wide scanning approaches identify candidate RNA elements but do not establish explicit cross-species correspondence. Methods specifically targeting lncRNA homology are most directly comparable: slnky infers lncRNA orthologs from whole-genome alignments using sequence-level conservation signatures, and lncHOME groups lncRNAs into families based on conserved genomic position combined with patterns of RNA-binding-protein motifs [40, 41]. Both rely on primary-sequence or motif-level signals to define homology. In contrast, CURIA does not require strong end-to-end sequence conservation within the ncRNA itself. It combines whole-genome-alignment-derived syntenic constraints with embedding-based similarity, so correspondence can be proposed for loci where conventional sequence- and motif-level similarity provides limited or incomplete evidence, without requiring curated families or explicit structural alignment.

4.3 Biological implications

Beyond the modular correspondence pattern noted above, two observations bear on interpretation. For island-branch loci, correspondence frequently concentrates in compact subregions within

broader transcript contexts in which CURIA detects substantially less localized embedding-supported correspondence. For short ncRNAs, although embedding-based similarity is strongly associated with sequence identity, the relationship is not uniform: at intermediate sequence identity ($\approx 40\text{--}60\%$), substantial variability is observed, indicating that loci with comparable sequence similarity can exhibit markedly different embedding-based similarity.

The island-based representation provides a practical way to capture this behavior: recurrent embedding-supported islands define candidate corresponding regions, while intervening regions show less detected localized correspondence. Correspondence patterns also vary across RNA classes: compact RNAs such as tRNAs and snoRNAs show strong and consistent signal, whereas lncRNA loci more often exhibit partial or fragmented correspondence [10, 12, 14]. Together, these observations support a model in which recurrent candidate elements are embedded within broader transcript contexts showing substantially less localized embedding-supported correspondence, emphasizing the importance of localized comparison in cross-species ncRNA analysis.

Given the heterogeneity of lncRNA evolution and annotation, a single universal orthology criterion may be premature. A more tractable near-term objective is to stratify candidate correspondence by the strength and type of comparative evidence. Recurrently matched candidate cores provide one such evidence layer, defining subsets that can be prioritized for subsequent functional and evolutionary investigation without treating recurrence alone as proof of orthology or function.

4.4 Limitations

This study has several limitations related to data availability, modeling assumptions, and evaluation. First, the lack of reliable ground truth for cross-species ncRNA correspondence remains a major challenge; existing annotations are incomplete and heterogeneous across species, and transcript boundaries—particularly for lncRNAs—are often ambiguous. As a result, evaluation based on annotation overlap provides only an approximate and conservative estimate of performance.

The recurrent-core screen also separates coverage from match quality. At the pipeline acceptance ceiling ($d \leq 0.10$), the 17-of-19 set contains 1,693 lncRNA loci, whereas requiring the same distance cutoff in every counted species reduces this to 170 loci at $d \leq 0.02$ and 21 at $d \leq 0.01$. These stricter subsets provide useful high-confidence candidates, but the distance is a length-dependent ranking statistic rather than a calibrated probability, so the cutoffs should be interpreted as sensitivity analyses rather than definitive biological quality classes.

Second, CURIA depends on the availability and quality of genome alignment chains. Because the search space is restricted to aligned regions, correspondence cannot be recovered outside chain coverage, and errors or fragmentation in alignments propagate directly to downstream predictions. In addition, the candidate-region classification step is adapted from protein-coding gene projection and may not fully capture the diversity of ncRNA correspondence patterns.

Third, the method relies on embedding representations derived from RNA foundation models. Although RiNALMo [26] provides a practical representation, it may not capture all relevant structural features. The island detector was trained on annotated compact ncRNAs enriched for structured RNA classes against genomic background. A genuine lncRNA core whose embedding pattern lies outside that positive class may therefore be classified as background and never reach the matcher. Repository held-out-family analyses show broad generalization across structured-RNA families but substantially weaker recall for lncRNA, consistent with this failure mode. In particular, CURIA is primarily sensitive to localized signal in embedding space and does not model long-range interactions or global RNA architecture. Because it assumes approximate collinearity within candidate regions and scores local similarity, it does not capture rearrangements of structured elements; loci lacking localized embedding signal are unlikely to yield detectable correspondence.

Fourth, CURIA does not establish transcriptional activity or molecular function; predicted regions represent candidate corresponding elements and require additional transcriptomic or experimental validation. Annotation-discordant calls have several competing interpretations—missing or imprecise query annotation, projection or assembly error, correspondence to a different nearby element, or a false-positive match—which cannot be resolved from interval overlap alone. We therefore report annotation agreement rather than treating it as either precision or recall.

Recurrent island matches may also include repeat-derived RNA segments, conserved DNA elements, or other sequence features that are not specific to RNA structure or function. Systematic stratification using RepeatMasker annotations is therefore a useful future extension, although this possible confounding does not by itself invalidate the observed matches.

Finally, CURIA does not explicitly distinguish between shared evolutionary origin and convergent similarity at the level of embedding space. While synteny strongly biases predictions toward corresponding loci with shared ancestry, embedding similarity may also capture convergent structural features or recurrent local motifs. Accordingly, high-confidence matches should be interpreted as candidate corresponding elements supported by both synteny and embedding similarity, rather than definitive evidence of shared evolutionary origin.

4.5 Embedding-based similarity and interpretability

Embedding representations used in CURIA are derived from RNA foundation models trained on large collections of RNA sequences without explicit structural supervision [16, 27, 28]. Prior work has shown that such models encode representations associated with RNA secondary structure and related functional properties, and can support downstream structure prediction tasks [42]. In this context, embedding similarity was strongly sequence-associated rather than independent of sequence. For compact short ncRNAs, MMD was largely redundant with the tested sequence metrics: its incremental predictive value was small overall and absent after excluding miRNAs. In the tested human–mouse and human–cow island samples, however, embedding-SW was not fully reducible to the tested sequence-derived metrics and added ranking information beyond nucleotide identity, local sequence alignment, k-mer similarity, composition, and length; the analysis does not establish that this additional information represents conserved RNA structure or function. Empirically, match scores form a continuous scale separating high-confidence matches from background signal. This interpretation is broadly consistent with observations from protein language models, where representations learned from sequence alone organize according to structural and functional properties [43].

Taken together, the sequence-baseline, recurrence, and folding analyses delimit what the embedding signal supports. In the tested island samples, synteny-constrained embedding similarity adds ranking information beyond the tested nucleotide metrics. The broad recurrent set is intentionally inclusive, whereas stricter per-species quorums define nested high-confidence subsets; no single embedding-SW distance cutoff is presented as biologically optimal. ViennaRNA agreement was heterogeneous, with moderate boundary-stable agreement for one MALAT1 core but weak or unstable agreement for the others. CURIA should therefore be interpreted as a correspondence framework using synteny and learned RNA representations, not as a direct predictor of structural conservation or function.

4.6 Future directions

The most informative next step is biological interpretation of the recovered cores. Predicted elements can be prioritized for secondary-structure analysis, candidate interaction surfaces and binding partners, transcriptomic support, and targeted validation. Comparing the number, order, and divergence of cores across species may reveal how lncRNAs gain, lose, or rearrange functional modules and may identify recurrent mechanisms that are obscured at whole-transcript scale.

Methodologically, combining core matches across species in a phylogenetic model could distinguish conserved ancestry from recurrent similarity and reduce the impact of a single poor assembly or chain set. Multi-scale representations may also connect locally detectable cores to longer-range RNA architecture.

Smith–Waterman alignment over the token-level cosine-similarity matrix was selected as a natural proof-of-concept objective because it localizes ordered embedding similarity while allowing gaps, but it is not assumed to be optimal, particularly because its accumulated score depends on both alignment length and per-position similarity. Possible future algorithmic extensions include length-normalized or explicitly length-regularized local alignment, soft-DTW or related order-aware differentiable objectives, optimal-transport-based matching, and local substring matching that minimizes MMD or another distribution distance subject to a minimum-length constraint or a regularization term preventing degenerate selection of very short substrings.

5 Data and code availability

The CURIA pipeline is available at the project repository: https://github.com/kirilenkobm/CURIA_pipeline. Preprocessed reference annotation files used in this study are provided in the repository under `input_data/reference_annotation/`. The recurrent-core tables described in Section 3 are included in the repository under `paper/recurrent_core_tables/`. This directory contains gene-level, core-level, and core-by-assembly match tables, distance and quorum sensitivity tables, and a README defining coordinates, identifiers, acceptance criteria, and derived fields. The historical gene-level summary remains available alongside the package. The per-species predictions and analysis tables underlying the preprint are provided under `preprint_results/`. The exploratory ViennaRNA supplement is provided as `paper/supplementary/vienna_structure_supplementary.pdf`, containing Supplementary Figures S1–S7 and Supplementary Table S1. Its machine-readable per-window results, source data, and diagnostic plots are provided under `analysis/vienna_structure_examples/`, including `malat1_windows.tsv`. Genome assemblies (2bit format) were obtained from NCBI [32] and UCSC Genome Browser [33] resources. Genome alignment chains were obtained from the TOGA project (Hiller lab) [1]: <https://genome.senckenberg.de/download/TOGA>. All external datasets are publicly available, and links to the exact sources are provided in the repository.

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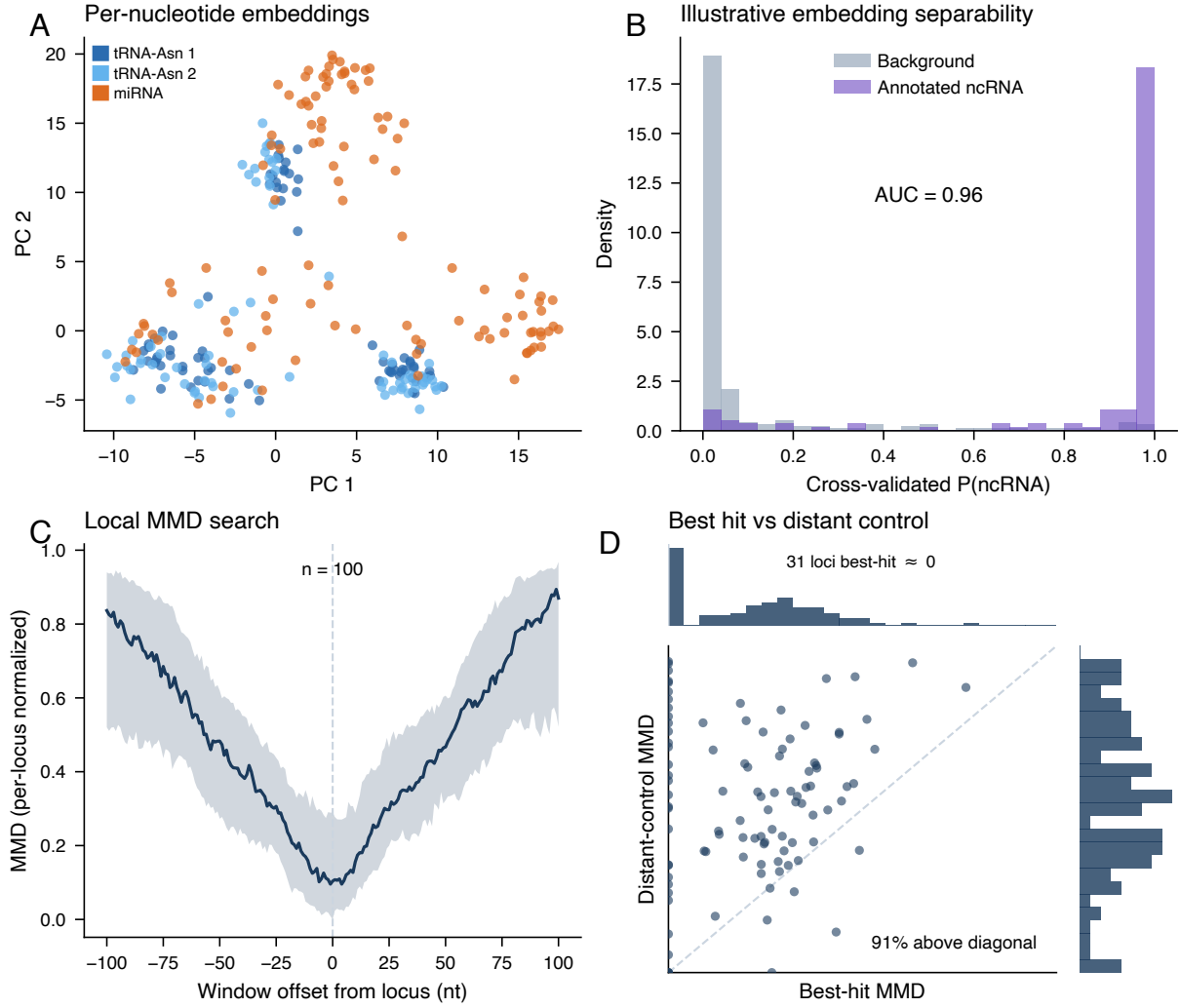


Figure 1: Illustrative embedding behavior supporting local ncRNA comparison. **(A)** Per-nucleotide embeddings (matching PCA, first two components) for two human tRNA-Asn-GTT gene copies (*tRNA-Asn-GTT-chr1-139* and *-140*, 74 nt each) and one pre-miRNA (*MIR103A1*, ENST00000362168, 110 nt), all from the human genome assembly hg38; each point is a nucleotide. The tRNA copies cluster together while the miRNA forms a distinct cluster. **(B)** Out-of-fold probability from a 5-fold cross-validated logistic classifier trained on the embeddings to distinguish annotated hg38 ncRNAs (purple; tRNA, snoRNA, miRNA) from background sequences (gray; intergenic and dinucleotide-shuffled): ncRNAs score near 1 and background near 0 (AUC ≈ 0.96), showing the embeddings separate the selected RNA-derived sequences from genomic background. Panel (B), titled “Illustrative embedding separability” within the figure, is an illustrative separability experiment on a small balanced set and is not the production island detector used in the pipeline (Section 2.7), which is trained on a different biotype set and negatives and operates at a different threshold. **(C)** Maximum Mean Discrepancy (MMD) between a short-ncRNA reference and windows tiled across its extended locus, versus offset from the annotated position (median and interquartile range over $n = 100$ short ncRNAs, each locus min-max normalized). MMD is minimised at the true position (offset 0) and rises with positional shift. **(D)** Best-hit MMD versus a distant control window (250 nt away) for the same $n = 100$ loci, with marginal histograms. Best-hit MMD concentrates near 0 (31 loci essentially perfect), whereas the distant control almost never does; 91% of loci lie above the diagonal (matched window more similar than the control).

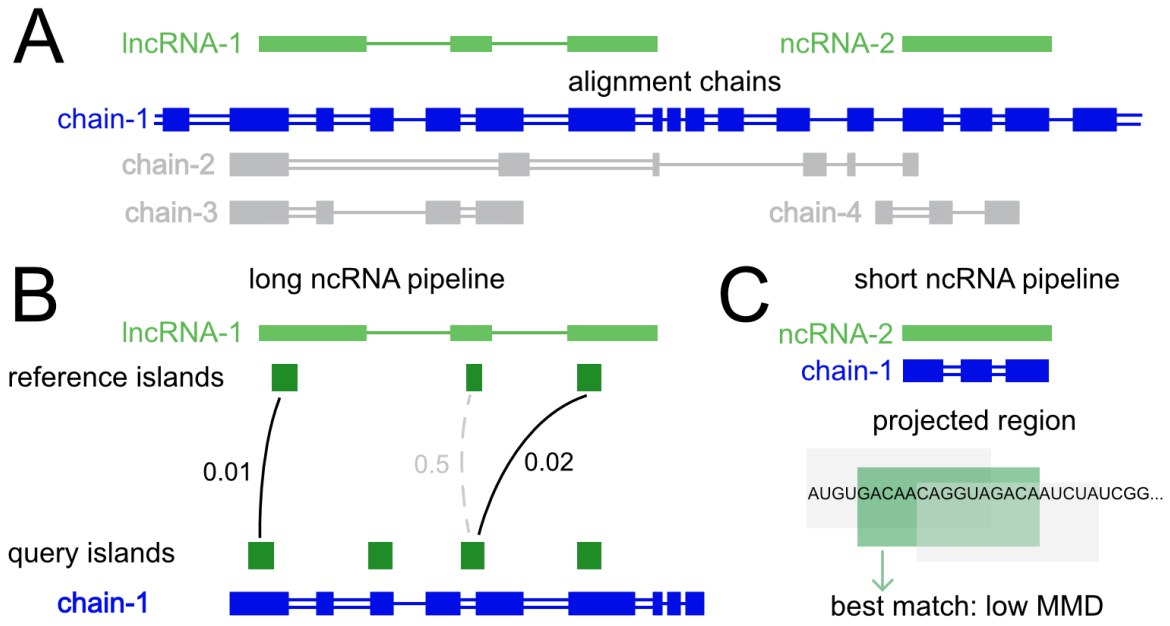


Figure 2: Overview of the CURIA pipeline for cross-species ncRNA correspondence analysis. **(A)** Genome alignment chains restrict the search space by defining candidate loci between reference and query genomes. **(B)** For island-branch loci, localized subregions (“islands”) are detected and matched across species using embedding-based similarity, enabling localization of candidate corresponding regions within transcripts. **(C)** For short ncRNAs, correspondence is resolved via local boundary refinement using sliding-window matching in embedding space.

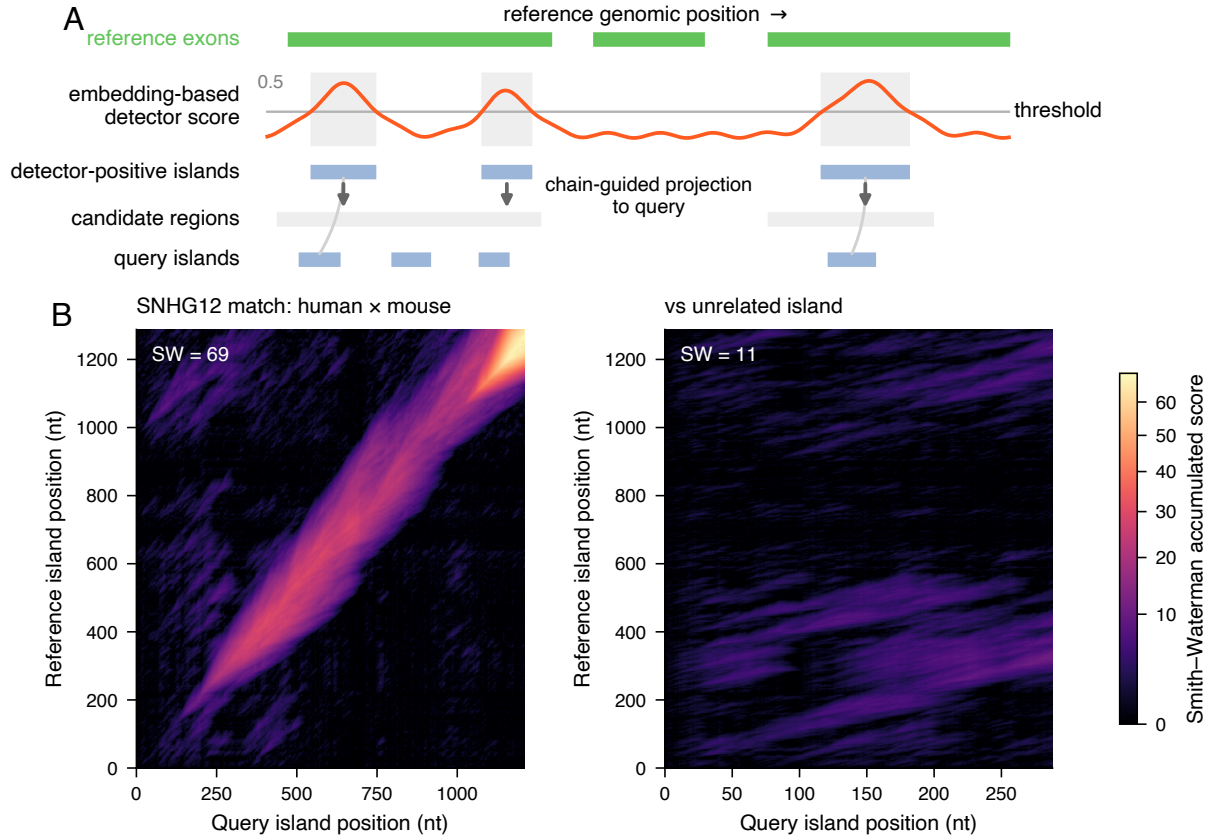


Figure 3: Identification and matching of localized ncRNA subregions using embedding-based similarity. **(A)** Schematic: an embedding-based detector score along a reference transcript is thresholded to define localized subregions (“islands”); islands are projected via genome alignment chains to candidate query regions, where corresponding islands are detected independently. **(B)** Island matching for a recurrent lncRNA region (SNHG12, human hg38 chr1 × mouse mm39 chr4). Each island is embedded once; the per-token cosine dotplot is aligned by Smith–Waterman, shown here as the accumulated-score matrix. The conserved matched region forms a single coherent diagonal ridge (embedding-SW alignment score 69), whereas the same human island against an unrelated mouse island yields no coherent alignment (embedding-SW alignment score 11).

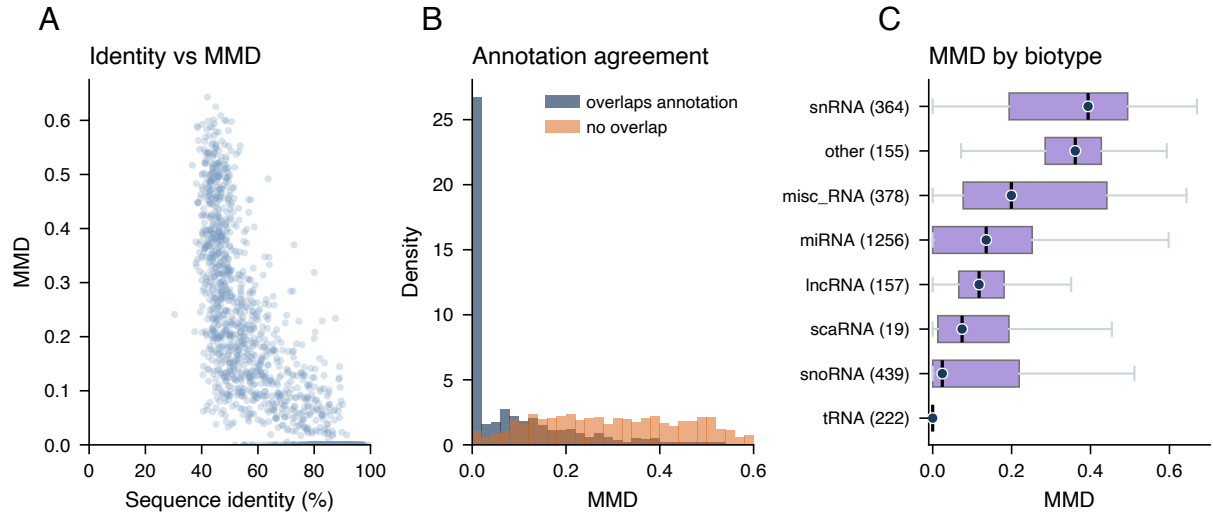


Figure 4: Behavior of embedding-based similarity (MMD) across short ncRNA loci for the human-mouse pair. **(A)** Sequence identity versus MMD for refined predicted loci. **(B)** MMD distributions for loci that do versus do not overlap mouse annotation: annotation-overlapping loci concentrate at low MMD, non-overlapping loci shift higher. **(C)** MMD distribution by ncRNA biotype, ordered by median.

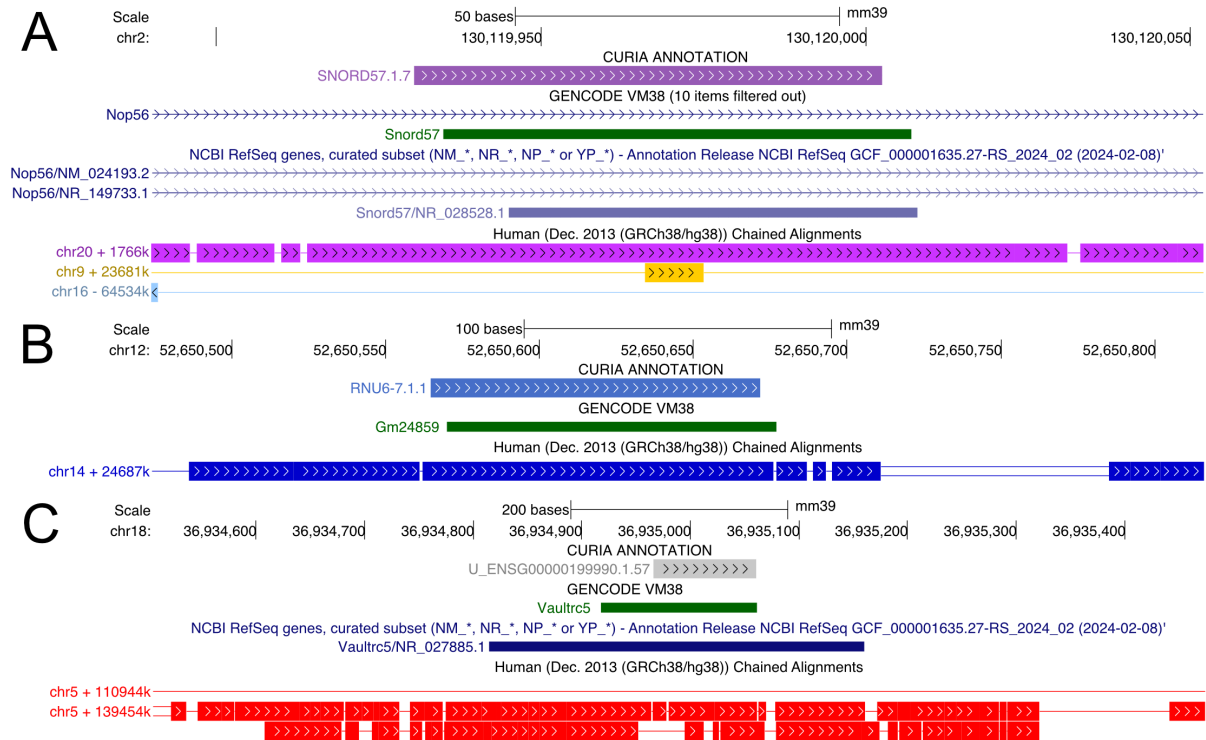


Figure 5: Recovery of short ncRNA loci across species using CURIA. Genome browser snapshots (UCSC) illustrating representative examples of short ncRNA projection from human (hg38) to mouse (mm39): a precisely localized SNORD57 match, RNU6-7 intersecting Gm24859, and a localized vault-RNA match with differing boundaries. SNORD57 has 74% edit-derived identity and reported MMD 0.00; RNU6-7 has 93% edit-derived identity and reported MMD 0.00; and the vault-RNA match overlapping Vault5 has 59% edit-derived identity, normalized nucleotide-SW score 0.35, and MMD 0.14.

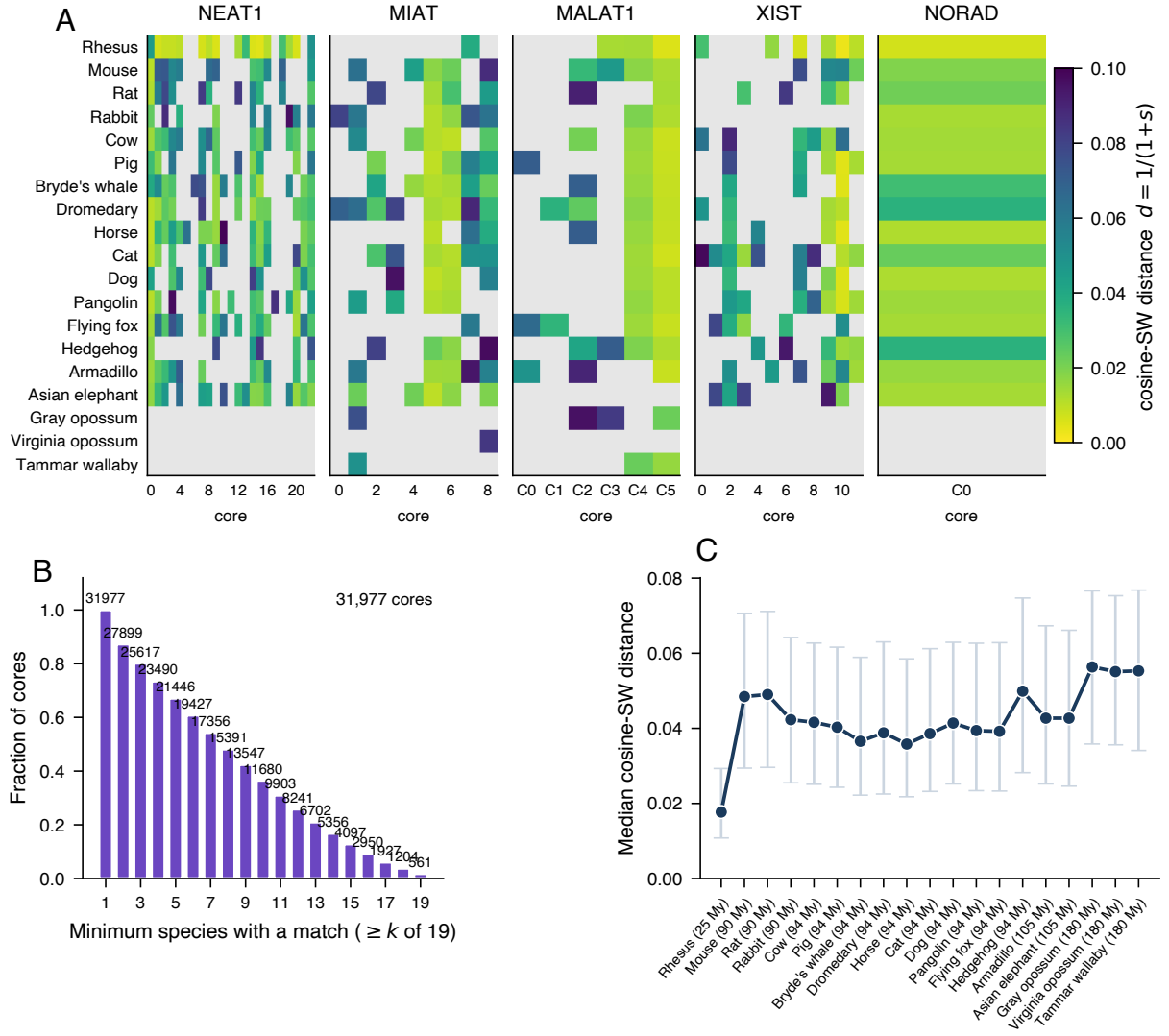


Figure 6: Modular and heterogeneous cross-species correspondence of localized regions. **(A)** Cross-species matching of core regions for representative lncRNAs (NEAT1, MIAT, MALAT1, XIST, NORAD); rows are species, columns are reference cores ordered along the transcript, color indicates embedding-SW distance $d = 1/(1+s)$. **(B)** Distribution of core reproducibility across species. **(C)** Median island-level match distance across species. This descriptive quantity is length-dependent and does not follow a strictly monotonic phylogenetic gradient. Panel (B) is conditional on the 31,977 aggregated reference-coordinate cores with at least one accepted match; loci without a detected island and islands without any accepted match are not in its denominator. This set contains 30,926 cores from loci annotated as lncRNA and 1,051 from other island-branch biotypes. Missing projection and failed matching both appear as an absent species in this panel.